Examining Network Commands in Linux

At the end of this episode, I will be able to:

1. Identify and explain common network command commands in Linux systems, given a scenario.

Learner Objective: *Identify and explain common network command commands in Linux systems, given a scenario*

Description: In this episode, the learner will examine common network troubleshooting commands used in Linux. We will explore command such as ping, ip, dig, mtr, nmap, tcpdump and more.

- What is a good command to use in Linux, when starting to troubleshooting connections?
    - *ip* - display and modify routing, network devices, interfaces and tunnels

      ```
      ip a/addr - display IP address
      ip a show eth0/lo - display information about the specified interface
      ip link show up - display all links with a state = UP
      ip route - displays IP routing table
      ```

    - *ping* - tests reachability of and endpoint or host

      ```
      ping 192.168.0.10 - sends ICMP echo requests to 192.168.0.10 to check connectivi
      ping -c 2 google.com - "-c 2": sends 4 echo requests to www.google.com. Useful f
      ping -s 90 google.com = -s defines size of ICMP packets in bytes
      ```

    - Process
        - Start with 127.0.0.1 (Loopback, verifies TCP/IP is functional)
        - Next, ping the local host's IP address (verifies local IP address)
        - Next, ping another host on the local network (verifies communication on the local network)
        - Next, ping the default gateway (verifies communication with the router, connected to the local network)
        - Finally, ping a remote host (verifies connectivity beyond the local network)
- What if we think name resolution is an issue, what command can we use?
    - *dig* - DNS lookup utility

      ```
      dig google.com - queries for DNS for an A record for google.com
      dig @8.8.8.8 google.com queries the specified DNS server for an A record
      dig -x 8.8.8.8 - performs reverse lookup
      dig +trace example.com - performs route tracing for DNS query
      dig example.com A AAAA +noall +answer - queries the DNS server for A and AAAA re
      ```

- If we would like to gather information about a connection path, how can be test this?
    - mtr - (My Traceroute) combines traceroute and ping functionality

      ```
      mtr google.com - performs traceroute, displaying packet path from source to dest
      mtr -c 10 google.com - set the maximum ping request to 10
      ```

- If we need to gather data off of the network for inspection, is there a command for this
    - *tcpdump* - dump/capture packets off of the network

      ```
      tcpdump -i eth0 - capture packets from specified interface
      tcpdump icmp
      tcpdump -i eth0 -w mycapture.pcap
        - capture from specified interface
      ```

```
- write packets to the mycapture.pcap file
```

- If we need to gather perform network scanning, what utilty can we use?
    - *nmap* - Network mapper, host discovery and port scanning

```
nmap 192.168.0.10 - Perform simple scan to identify open ports and services
nmap myserver.com - Perform simple scan to identify open ports and services
nmap
nmap 192.168.0.1-50
```